

COURSE PLAN

I Trimester 2026-27

SECTION I

Trimester	I	Programme & Class	1 MCA A & B
Course Code	MCA413-1	Course Type	Core (Theory+Practical)
Course Title	Full Stack Development	LCA	No
Total Hours	75	Hours per week	3+4=7
Faculty Name	Dr.Cynthia T Dr. Vijay Arputharaj J	Contact Details and Office Hours	cynthia.t@christuniversity.in vijay.arputharaj@christuniversity.in
Class Policies and Guidelines	● Students are required to maintain high standards of conduct while attending the classes and in all correspondence. ● Complete the class assignments on time and submit the same for the evaluation. ● Students who are not able to attend a class due to unavoidable reasons, are expected to inform the faculty handling the session/class teacher via official email, along with reason for absence. ● Dress code mandated by the department to be followed. ● Students are expected to be inside the classroom as per the timeslot mentioned in the time table. Attendance will be taken within the first 5 minutes. Late comers will not be entertained for attendance. ● Usage of Laptop / mobile phone / Headphones inside the class is strictly prohibited. However, during the demo / practice sessions students will be permitted to use their laptop. ● Strictly adhere to the class policies of the University and Computer Science Department ● Active and voluntary participation is expected during teachings and discussions ● All the information regarding this course will be communicated through LMS		

Course Description	The Full Stack Development course is designed to equip students with the skills and knowledge required to build modern, scalable web applications. The curriculum covers HTML5, CSS3, JavaScript, React.js, Node.js, Express.js, MongoDB, and Next.js, providing hands-on experience in both front-end and back-end development. Students will learn to design interactive user interfaces, manage databases effectively, and develop robust server-side applications. By the end of the course, learners will be able to integrate these technologies to create dynamic, full stack solutions aligned with industry standards.			
Course Objectives (Optional)	This course is designed to gain proficiency in full stack development, acquire the skills to build dynamic and robust web applications using advanced technologies, and develop strong web programming practices. The course also enables learners to apply their knowledge to solve real-world challenges in the field of web development.			
Course Outcomes	CO1: Apply HTML5 semantic features, GIT for version control and Tailwind CSS to develop structured and maintainable websites. CO2: Implement client-side scripting using JavaScript for HTML5 APIs, forms, events, browser features, and JSON-based data exchange. CO3: Develop dynamic single-page applications using React.js with hooks, routing, Axios, and state management. CO4: Design and optimize Node.js applications integrated with MongoDB to perform efficient CRUD operations and manage data effectively. CO5: Build high-performance web applications using Next.js with SSR and SSG for improved performance and SEO.			
SECTION II				
Week	Module/ Unit/ Topic details	Andragogy	Ungraded formative Assessments	Resource/ Reference details
Week 01 (1/6/26 - 6/6/26)	Unit 1: HTML5 New Elements – HTML5 Semantics –Canvas – HTML Media – Git-commit –rollback – remote repository – GitHub – merge conflict	Project-Based Learning	Code Reviews (Peer based)	HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery), DT Editorial Services, Dreamtech Press 6th Edition, 2023. (Module1)
Week 02 (8/6/26 - 13/6/26)	Unit-1: CSS specificity rule – Pseudo selectors – media queries –	Portfolio Development	Real time collaboration	HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery), DT

	flexbox – responsive web design – transition Lab Exercises: Develop a responsive webpage layout using Tailwind CSS. Include a header, navigation bar, main content, and footer and Tailwind Typography plugin.			Editorial Services, Dreamtech Press 6th Edition, 2023. (Module1)
Week 03 (15/6/26 - 19/6/26)	Unit-1: Tailwind responsive grid – Components (Navbar, tables, heroes, carousel, modal etc.,) – font awesome icons. Introduction to Core JavaScript - Use JavaScript to interact with some of the new HTML5 APIs Lab Exercises: Demonstrate HTML5 APIs Unit-2: CLIENT-SIDE SCRIPTING JavaScript objects - JS Forms	Scaffolded Learning Path	CIA -1A : MCQs CIA -1B : Practical Questions	HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery), DT Editorial Services, Dreamtech Press 6th Edition, 2023. (Module1)
Week 04 (22/6/26 - 27/6/26)	Unit-2: Events and Event handling- Async await- JS Navigator-JS Cookies Lab Exercises: Demonstrate event handling.	Flipped Classroom	Debugging Challenges	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 05 (29/6/26 - 4/7/26)	Unit-2: Introduction to JSON-JSON vs XML-JSON Objects-fetch API Lab Exercises: Create a web application using vanilla JavaScript with a database.	Peer Teaching / Code Pairing	Mini code sprints	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 06 (6/7/26 - 11/7/26)	Unit-3: React.JS : Package Manager	Peer Teaching / Code Pairing	CIA -II (Practical Exam)	Modern Full-Stack Development: Using

	(NPM) - ES6- Introduction to React.js - Create React App & React file structure Lab Exercises: Create a React application that allows users to enter details through a form and display the profile dynamically after submission.			TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 07 (13/7/26 - 17/7/26) Mid Trimester Examination	Unit-3- JSX and Components -passing and destructuring props - React Hooks – Axios - Images and Forms-Conditional Rendering - Routes - Redux Lab Exercises: Develop SPA (Single Page Application) with React.js and Implement CRUD Operation using React.js.	Problem-Based Learning	Bug hunts	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 08 (20/7/26 - 25/7/26)	Unit-4-Node JS and MongoDB Introduction to Node.js. Express JS - Node mailer - Mongo DB Fundamentals - Documents and Collections Lab Exercises: Demonstrate Node.js file system module.	Problem-Based Learning	Hackathons	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 09 (27/7/26 - 1/8/26)	Unit-4: Databases, Query Language, MongoDB CRUD Operations (Create, Read, Projection, Update, Delete) - Connecting Node.js with MongoDB	Problem-Based Learning	API Practice	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.

Week 10 (3/8/26 - 8/8/26)	Unit-4: Building simple RESTful services using Express and MongoDB Lab Exercises: Build simple RESTful services using Express and MongoDB	Case Study	CIA-III Theory exam	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 11 (10/8/26 - 14/8/26)	Unit-5: NEXT.JS Setting up Environment-initiating new Next.js Project-Routing in Next.js-Server and Client Components, Data Fetching Methods - Data Fetching Methods using React Query	Gamification Elements	Short Explainer Videos	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 12 (17/8/26 - 22/8/26)	Unit-5: CSS in Next.js - Integrating Tailwind CSS- Mobile-First Design for responsive and mobile-first design Lab Exercises: Implement server-side rendering and client-side rendering in Next.js. Display data fetched from an API using both methods. Use Tailwind CSS with Next.js to create a mobile-first, responsive webpage layout.	Mentorship & Industry Interaction	CIA-IV – Concluding marks	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRES, 2 nd Edition, 2022.
Week 13 (24/8/26 - 29/8/26)	Revision and Marks consolidation	Revision	ETE- Lab exam	Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker, Frank Zammetti, APRESS, 2 nd Edition, 2022.

SECTION III

Mapping of needs with the syllabus:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	1	2	2	-	2	3
CO2	2	3	2	2	2	3	-	2
CO3	1	2	1	1	2	-	2	1
CO4	2	2	3	2	2	2	2	-
CO5	2	2	3	2	2	2	-	1

Course Outcomes	Local, Regional, National, Global and Development Needs (LRNG)	Cross-cutting issues relating to Professional Ethics, Gender, Human Values, Environment and Sustainability	Employability/Entrepreneurship/ Skill Development	Sustainable Development Goals
CO1	National and Global	Professional Ethics, Environment and Sustainability	Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
CO2	Regional, National and Global	Professional Ethics, Environment and Sustainability, Gender equality	Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
CO3	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Employability/Entrepreneurship/ Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
CO4	Regional, National and Global	Professional Ethics, Environment and Sustainability, Gender equality	Employability/Entrepreneurship/ Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
CO5	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Employability/Entrepreneurship/ Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
Topic/Module Details				

I . Overview of Web Technologies and HTML5	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
II .Client-Side Scripting	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
III . React JS	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Employability/Entrepreneurship/ Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
IV Node JS and MySQL	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Employability/Entrepreneurship/ Skill Development	SDG 4, SDG 9, SDG 13, SDG 17
V.Next JS	LRNG	Professional Ethics, Environment and Sustainability, Gender equality	Employability/Entrepreneurship/ Skill Development	SDG 4, SDG 9, SDG 13, SDG 17

A template to map the Course Outcomes against the components of the assessment is given below:

Course Outcomes of the course	Components of Assessment (Marks)				
	CIA I	CIA II	CIA III	CIA IV Overall	ETE
CO1	10	20	-	10	5
CO2	15	20	-	10	5
CO3	-	10	10	10	10
CO4	-	-	10	10	15
CO5	-	-	5	10	15

Assessment outline:

	Components of assessment
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Components	CIA I	CIA II	CIA III	CIA IV	ETE	Attendance
Marks	25	50	25	50	50	5
Theory (50% CIA) CIA II (50) + CIA IV (50) = 100 Marks Will be downscaled to 45 Marks Attendance: 5 Marks, Total: 45+5 = 50 Marks			Practical (50% ETE) CIA I (25) + CIA III (25) + ETE (50) = 100 Marks Will be downscaled to 50 Marks			

SECTION IV

Assignment Component:: CIA 1 will comprise two components, each evaluated out of 10 and 15 marks.

CIA I component-1

Assessment Description	MCQ
Individual Assignment Details (Component -I/ Component-II/ Component- III)	CIA - Component -IA MCQ Test – 10 Marks
Date of Examination	Week 3
Assignment Description	Portion: Unit-I, Unit-2 (Half) No. of questions: 10-15 equally weighted questions (0.5-1 Mark) Marks: 10 No Negative marks Time Duration: 15 Minutes Mode of Test: Online (LMS)
Learning Outcomes	LO1: Identify suitable HTML5 semantic structures and responsive design techniques for modern web development scenarios. LO2: Analyze problem scenarios to select appropriate CSS, Tailwind CSS, and front-end implementation strategies. LO3: Apply Git-based version control practices for efficient code management, collaboration, and deployment workflows.

Evaluation Rubrics		
Rubrics	Score	Impression
R1	8-10	Proficient
R2	5.5-7.5	Good
R3	4-5	Satisfactory
R4	0-3.5	Poor

CIA I component-2

Assessment Description	Problem solving
Individual Assignment Details (Component -I/ Component-II/ Component- III)	CIA - Component -IB CIA - Component -IB (Problem Solving and Prototyping) Problem solving + Prototyping + Documentation – 15 Marks
Date of Examination	Week 3
Assignment Description	Develop a prototype-based solution for a real-world problem using Unit I & II concepts, accompanied by documentation. Responsible AI may be used for self-learning, debugging, and solution enhancement. Marks: 15 Due date: 7 days after posting assignment Mode of Submission: Online (LMS) and Presentation
Learning Outcomes	LO1: Analyze a real-world or domain-specific problem and identify appropriate client-side scripting solutions using HTML5, CSS, and JavaScript concepts. LO2: Develop an interactive prototype implementing forms, events, validation, browser features, and JSON-based data handling to address the identified problem. LO3: Document the solution design, implementation approach, and responsible use of AI tools for self-learning, debugging, and solution enhancement.

Evaluation Rubrics		
Rubrics	Component	Marks
R0	MCQ Answers (Equally weighted questions)	10
R1	Problem definition & innovation	5
R2	JavaScript implementation (events/forms/fetch/JSON)	5
R3	Documentation + GitHub quality	5

CIA – Component 2:

Assessment Description	MSE (Centralized Theory Written Test)
Individual Assignment Details (Component -I/ Component-II/ Component-III)	CIA - Component -II Theory Written Test Mid Semester Exam
Date of Examination	Week-7
Assignment Description	Portion: Unit-I, Unit-II, Unit-III (Half) No of questions: 5 - Last 2 questions with internal choices Each question is for 10 Marks Total Marks: 50 marks Time Duration: 2 Hours Mode of Test: Offline
Learning Outcomes	LO1: Interpret front-end development scenarios and justify the use of HTML5, CSS/Tailwind, and Git workflows in web application design. LO2: Analyze client-side scripting problems and propose appropriate JavaScript-based solutions using events, asynchronous operations, forms, and data exchange mechanisms. LO3: Examine React.js application requirements and recommend suitable component-based design and state management strategies for SPA development.

Evaluation Rubrics		
Rubrics	Score	Impression
R1	38-50	Proficient
R2	30-37.5	Good

R3	20-29.5	Satisfactory
R4	0-19.5	Poor

CIA - Component 3:

Assessment Description	Creative Designing
Individual Assignment Details (Component -I/Component-II/Component-III)	CIA Component -- III Practical Test
Date of Examination	Week-9
Assignment Description	Portion: Unit I, Unit-II , Unit-III, Unit-IV(Half) Duration: 2 Hours Total Marks: 25 marks for Practical test No of questions: One Scenario based question Mode of Submission: Online (GitHub and in LMS)
Learning Outcomes	LO1: Design and implement dynamic front-end interfaces using React.js with component-based architecture, state management, and event-driven interactions. LO2: Build and test full-stack functionality by integrating Node.js, Express.js, and MongoDB for CRUD-based application development. LO3: Develop a functional Next.js prototype demonstrating routing, rendering concepts, and responsive web design practices.

Evaluation Rubrics for Component (Practical Test)		
Rubrics	Components	Marks
R1	Understanding of Problem statement and Requirements	5
R2	Application of Concepts from the Syllabus (Checklist based evaluation)	10
R3	Code Design and Structure + Code Comments (2) Innovation / UI/UX / Responsiveness+ Validations (3)	5
R4	Functionality and Correct Output	5

	Problem Solving and Logical Reasoning (2) + Viva (3)	
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CIA- Component 4

Assessment Description	Dynamic Web page designing skills
Individual Assignment Details (Component -I/ Component-II/ Component-III)	CIA - Component -IV Regular Lab exercises evaluations
Assignment Description	Portion: 10 Lab Exercises Total Marks: 100 marks (Converted to 30 Marks) Mode of Submission: Online (GitHub and in LMS)
Learning Outcomes	LO1: Develop structured, responsive, and maintainable web interfaces using HTML5 semantic elements, CSS/Tailwind CSS, JavaScript, and browser-supported features. LO2: Implement interactive client-side functionalities using JavaScript forms, events, asynchronous programming, JSON, and API-based data handling. LO3: Design dynamic web applications using React.js, component-based architecture, hooks, routing, and state management techniques. LO4: Build full-stack application functionalities using Node.js, Express.js, MongoDB, and RESTful CRUD operations. LO5: Develop responsive modern web applications using Next.js concepts including routing, rendering, API integration, and deployment-ready front-end practices.

Evaluation Rubrics		
Each program carries 10 marks and evaluation rubrics for each program would be:		
Rubrics	Evaluation Criteria	Marks
R1	Execution/ Implementation	2 Marks
R2	Concept Clarity	2 Marks
R3	Formatting and Validation	2 Marks
R4	Complexity/Self Learning	2 Marks
R5	Github Collaboration	2 Marks

ETE- Component 5

Assessment Description	Evaluate Full stack design and development
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Individual Assignment Details (Component -I/ Component-II/ Component-III)	ETE - Component -V Lab exam
Date of Submission	Week-12
Ass Description	Portion: Unit 1, Unit 2, Unit 3, Unit 4, Unit 5 Total Marks: 50 marks. Question: Domain based web application from any domain with 3/5 pages (HTML5, React JS,Next JS, CSS, Java script, Database Connectivity, Form validation) Mode of Submission: Website developed should be posted in GitHub/LMS Duration : 2 Hours
Learning Outcomes	LO1: Design structured, accessible, and responsive web interfaces using HTML5 semantic elements, CSS/Tailwind CSS, and client-side design principles. LO2: Implement interactive client-side functionalities using JavaScript forms, event handling, browser features, asynchronous programming, and JSON-based data exchange. LO3: Develop dynamic front-end application modules using React.js concepts such as components, hooks, routing, and state management. LO4: Build and integrate server-side application logic with Node.js, Express.js, MongoDB, and RESTful CRUD operations. LO5: Construct a complete modern web application using Next.js with responsive design, routing, rendering concepts, and full-stack integration.

Evaluation Rubrics for Component (Practical Test + Written Component)		
Rubrics	Components	Marks
R1	Understanding of Requirements (Feature sketching)	5
R2	Application of Concepts from the Syllabus (Checklist-based evaluation)	25
R3	Code Design and Structure(2) + Code Comments (2) Innovation in UI/UX (2) / Responsiveness (1) + Validations (3)	10
R4	Functionality and Correct Output+ Problem Solving and Logical Reasoning (5)	10

	Github (Submission and Readme) (2) + Viva (3)	
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COURSE SPECIFIC AI TOOLS AND RESPONSIBLE AI USAGE GUIDELINES

A. AI Tools for Teaching the Course (Only for Faculties)

Unit/ Topic	Tool	Context
HTML/CSS/Tailwind Teaching Sample Activity: Generate a responsive navbar using Tailwind and ask students to improve accessibility.	ChatGPT	- Generating sample HTML structures - Explaining CSS specificity - Responsive layout examples - Tailwind component generation
	CodePen AI / Frontend AI tools	- live HTML/CSS experimentation - visual layout testing
JavaScript Teaching Sample: Generate a JSON fetch example and explain line by line.	GitHub Copilot	- Explaining JS syntax - Event handling examples - Async/await demos - Fetch API examples
	ExplainDev	- Explaining existing JS code - Breaking down browser APIs
React / Next.js Teaching	v0 by Vercel	- Rapid prototyping of React/Next.js interfaces for learning component structure - React component scaffolding - Next.js layout examples
	ChatGPT	- Hooks explanation - Routing examples - State management concepts
Backend (Node.js / Express)	GitHub Copilot	- REST API scaffolding - Middleware examples - Route generation
	Postman AI Assistant	API testing API testing, request generation, endpoint documentation assistance
MongoDB and Debugging During Lab	MongoDB Atlas AI Assistant	- Query generation - Aggregation explanation - Schema suggestions

	ChatGPT / Phind	• To crosscheck debug errors • To understand stack traces • To compare code approaches
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B. AI Tools for Assessment & Evaluation (Only for Faculties)

Component	Tool	Context
Practical Viva Preparation	ChatGPT	• To prepare viva questions • To get scenario-based questions • To collect debugging questions
Coding Assessment	HackerRank / CodeSignal AI-assisted evaluation	• Timed coding tests • JS problem solving • API logic tasks
Assignment Rubric Generation and Rigor setting	ChatGPT	• Rubrics for mini project • REST API assignment • MongoDB CRUD work
Plagiarism / Code Similarity	MOSS, Codequiry, Turnitin/Drillbit/ iThenticate	• Plagiarism reports • AI/Similarity reports • Code similarities

C. Student Engagement Using AI (Only for Faculties)

AI Code Challenges	Prompt Engineering for developer	AI vs Human Debug Battles	AI Design-to-Code Challenges	AI API Documentation	AI Peer Review
Students complete/refactor AI-generated code	Students compare prompts for better API/UI generation	Compare manual vs AI debugging	Convert AI UI suggestions into working apps	Generate API docs and validate accuracy	Peer verifies AI-generated suggestions

D. Responsible AI Use Worksheet – Full Stack Development (Only for Students)

Student Name:	
Course code and course name:	
Section 1: AI Tool Used	Mention tools used:

	☐ ChatGPT ☐ GitHub Copilot ☐ v0 ☐ MongoDB AI Assistant ☐ Postman AI ☐ Other _____
Section 2: Purpose	Mention the purpose of use: ☐ Understanding concepts ☐ Debugging ☐ Code generation ☐ UI design ☐ API documentation ☐ Query generation ☐ Assignment support ☐ Other _____
Section 3: Prompt Used	Mention the prompts used:
Section 4: AI Output Verification	AI output verification and how?
Section 5: Reflection	1. What part of the AI output was helpful? 2. What mistakes did AI make? 3. What did you modify manually? 4. What did you learn without depending entirely on AI?
Section 6: Ethical Commitment and Responsible use	☐ I understand AI suggestions may be incorrect. ☐ I tested all generated code myself. ☐ I did not submit AI-generated code without understanding it. ☐ I cited AI use where required.
Suggested AI Policy for Full Stack Development course:	
Allowed • concept clarification • debugging assistance • code explanation • UI prototyping	Not allowed • Submitting AI-generated work without comprehension, validation, or attribution • copied AI-generated project work • fabricated documentation